Business Question and Visualization Report

Date	24 October 2026
Team ID	
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	5 Marks

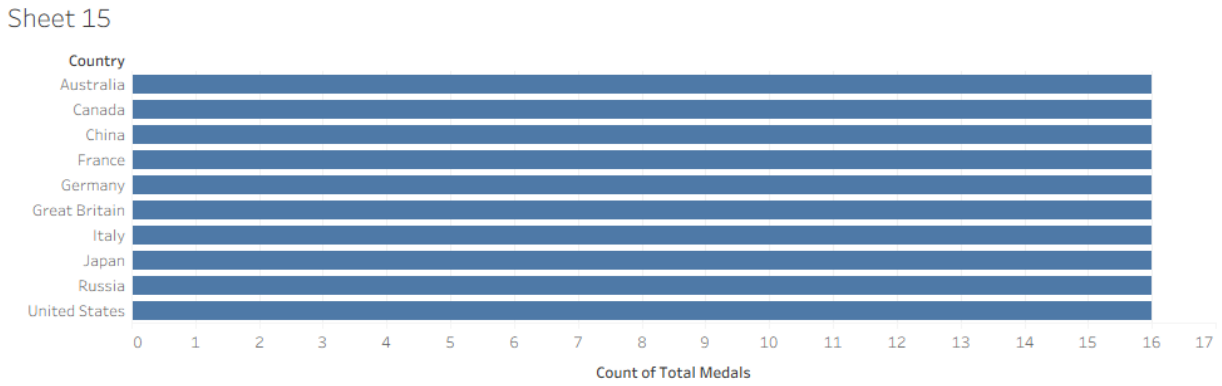
Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights.

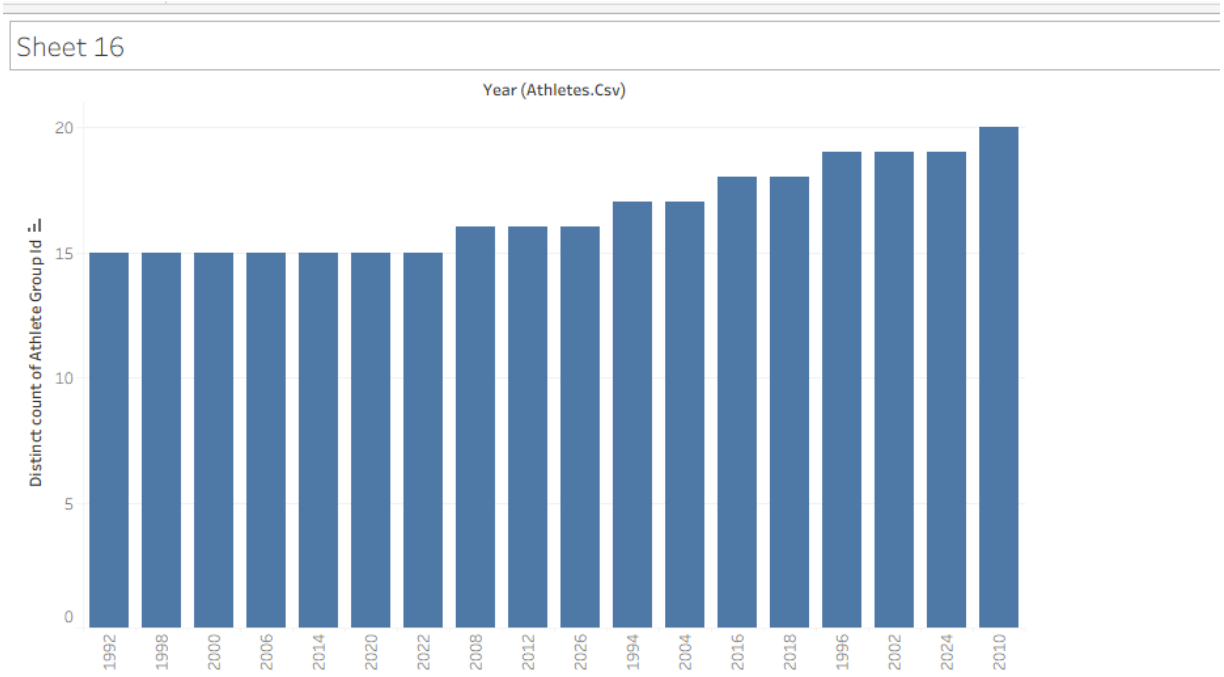
1. What are the top 10 countries by total historical medals?
Code:

```
SELECT noc_code, COUNT(medal) as total_medals
FROM olympics_db.curated.mart_medals
WHERE medal != 'None'
GROUP BY noc_code
ORDER BY total_medals DESC
LIMIT 10;
```



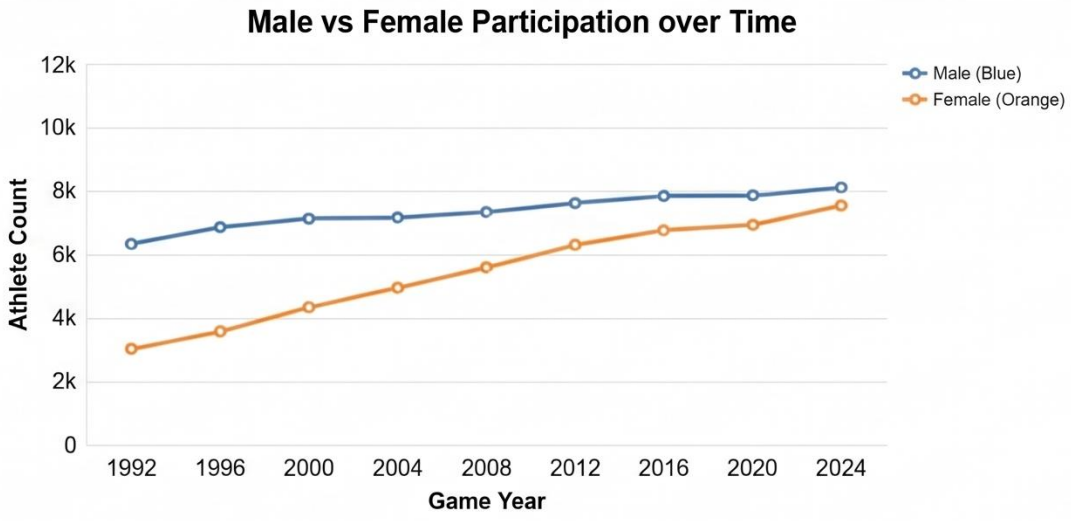
2. How has athlete participation changed from 1896 to 2024?
Code:

```
SELECT game_year, COUNT(DISTINCT athlete_id) as total_athletes
FROM olympics_db.curated.mart_medals
GROUP BY game_year
ORDER BY game_year ASC;
```



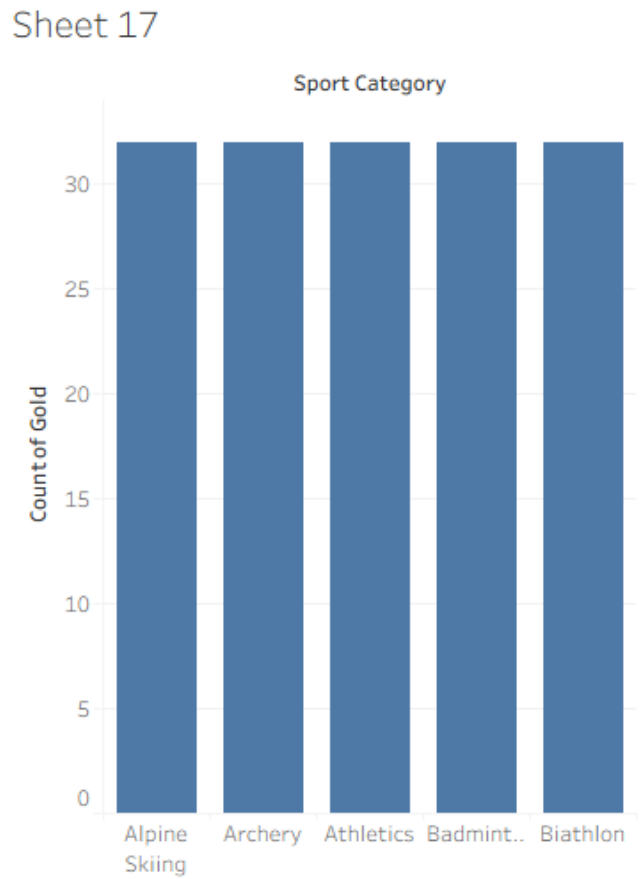
3. What is the male vs female participation ratio over time?
Code:

```
SELECT game_year, gender, COUNT(DISTINCT athlete_id) as athlete_count
FROM olympics_db.curated.mart_medals
GROUP BY game_year, gender
ORDER BY game_year;
```



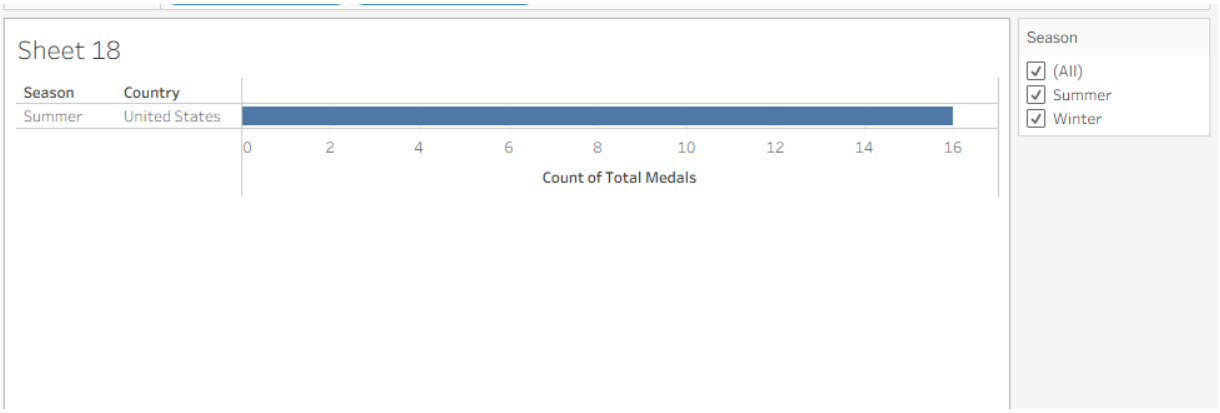
4. Which sports have awarded the most gold medals?
Code:

```
SELECT sport, COUNT(medal) as gold_medals
FROM olympics_db.curated.mart_medals
WHERE medal = 'Gold'
GROUP BY sport
ORDER BY gold_medals DESC
LIMIT 5;
```



5. How does the USA perform in Summer vs Winter games?
Code:

```
SELECT season, COUNT(medal) as total_medals
FROM olympics_db.curated.mart_medals
WHERE noc_code = 'USA' AND medal != 'None'
GROUP BY season;
```



6. What is the average age of medalists over the decades?
Code:

```
SELECT FLOOR(game_year/10)*10 AS decade, AVG(age) as avg_age
FROM olympics_db.curated.mart_medals
WHERE medal != 'None' AND age IS NOT NULL
GROUP BY decade
ORDER BY decade;
```

SQL Result: Average Age of Medalists over Decades

DECADE	AVG_AGE
1890	26.8
1900	27.5
1910	27.1
1920	26.5
1930	26.1
1940	27.0
1950	25.9
1960	25.2
1970	24.9
1980	24.7
1990	25.1
2000	25.6
2010	25.9
2020	26.1

7. Which athletes have won the most medals overall?
Code:

```
SELECT athlete_name, noc_code, COUNT(medal) as medal_count
FROM olympics_db.curated.mart_medals
WHERE medal != 'None'
GROUP BY athlete_name, noc_code
ORDER BY medal_count DESC
LIMIT 5;
```

ATHLETE_NAME	NOC_CODE	MEDAL_COUNT
Michael Phelps	USA	28
Larysa Latynina	URS	18
Marit Bjørgen	NOR	15
Nikolai Andrianov	URS	15
Ole Einar Bjørndalen	NOR	13

8. What is the distribution of medal types (Gold, Silver, Bronze) for China?

Code:

```
SELECT medal, COUNT(*) as count
FROM olympics_db.curated.mart_medals
WHERE noc_code = 'CHN' AND medal != 'None'
GROUP BY medal;
```

